

ANSHUL JAGTAP

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EDUCATION

University of California - Santa Cruz

September 2022 - August 2025

Bachelor's, Computer Science

GPA: 3.72

- Data Structures & Algorithms, Machine Learning, Artificial Intelligence, NLP, Discrete Math, Statistics and Probability, Principles of System Design, Introduction to Management of Technology I, Calculus, Multivariate Calculus

PROFESSIONAL EXPERIENCE

Baskin Engineering

Santa Cruz, CA, USA

Undergraduate Researcher

February 2024 - Present

- Developing distributed control algorithms using Python and MATLAB for campus eVTOL simulations, improving path efficiency by 25% and reducing energy consumption.
- Collaborated with IIT Madras to optimize atmospheric sensing models, enhancing air-ground data collection safety by 20%.

Akhetonics

Berlin, Germany

Software Engineer Intern

June 2024 - August 2024

- Developed and optimized Graph Coloring algorithms in C# using .NET for Sapphire Compiler, reducing register allocation time by 30%, which accelerated build processes across the development team.
- Enhanced memory management by implementing advanced NUnit test cases for graphs with 100,000+ registers, leading to a 60% improvement in time complexity
- Translated SPIR-V Bit Instructions to Ray Core for 16-bit ISA, streamlining GPU performance.

PROJECTS

Multi-Threaded HTTP Server

November 2024 - December 2024

Developer

- Developed a multi-threaded HTTP server in C using a fixed thread pool and thread-safe queue to handle concurrent client requests, Improved throughput by 4× over single-threaded implementation
- Engineered a synchronization-safe audit logging system using pthread_mutex to ensure correct request ordering and durability of logs under concurrent access, in compliance with strict linearizability requirements.
- Implemented a per-resource locking scheme by designing a hash table to map URIs to rwlock_t locks, reducing unnecessary synchronization and significantly improving server throughput while maintaining correctness under race conditions.

Trading Strategy Simulator: - [Link to project](#)

July 2024 - August 2024

Developer

- Developed a full-stack stock market analysis platform in Streamlit integrating multiple technical indicators to deliver buy/sell/hold recommendations with up to 85% historical backtest accuracy.
- Implemented secure user accounts and dashboards for 50+ users, enabling favorites, analysis history, and admin analytics tracking 1,000+ analyses.
- Integrated Yahoo Finance API with interactive Plotly charts, achieving sub-2s average load times for global stock analysis on desktop and mobile with a responsive UI.

ASL Gesture Detection Software - [Link to project](#)

November 2023 - December 2023

Developer

- Engineered a real-time ASL detection model using TensorFlow/Keras, trained on a custom dataset of 5,000+ labeled gesture images, achieving 95% accuracy in diverse lighting conditions.
- Optimized real-time performance using OpenCV and cvzone, achieving gesture recognition speeds of <50ms per frame, enabling seamless user interaction.
- Enhanced model robustness through data augmentation techniques, including rotation, scaling, and background noise reduction.

SKILLS

Technical Skills: MATLAB, Python, TypeScript, Kotlin, SQL, JavaScript, HTML/CSS, C/C++, C#, Java, R, Machine Learning, AI, Gitlab, Github, Git, Assembly, HTML, CSS, React.js, REST APIs, Fintech, Linux/Unix, JavaScript, RISC-V, Next.js, AWS

Financial Skills: Tradingview, Financial Modeling, Fundamental Analysis, Excel/Numbers/Sheets, Adobe Premiere Pro